



Modelling Financial Markets

Seminar 17: Bubbles and Crashes — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BIC: Bayesian Information Criterion
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ EM: Expectation-Maximisation (algorithm)
- ▣ FCLT: Functional Central Limit Theorem
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ LR: Likelihood Ratio (test)
- ▣ MC: Monte Carlo (simulation)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Separate fundamental value and bubble in a present-value model
- ▣ Derive the Dickey–Fuller and mildly explosive limits; run GSADF with simulated and bootstrap critical values and measure its size
- ▣ Test hit rates and the number of regimes with dependence-robust or bootstrap inference
- ▣ Fit an LPPLS model and judge how stable its critical time is
- ▣ Estimate a Markov-switching model and read filtered and smoothed probabilities

- ▣ Background: Lecture 17; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** ten computational exercises with inference (B1–B10)
- ▣ **Part C:** one open question, a seed for the team project (C1), and the critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Explosiveness tests: log levels, no lagged terms, minimum window $r_0 = 0.01 + 1.8/\sqrt{T}$, minimum duration $\lceil \ln T \rceil$ periods
- Always ask: would this signal have been available at the time, and in which direction did prices move?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (solving the pricing equation forward), A3 (Dickey–Fuller limit), A5 (Markov switching: durations and one filter step), A7 (LPPLS: linear parameters and filter)
- Part B: B1 (GSADF on the price-dividend ratio), B3 (Bitcoin: Monte Carlo vs wild bootstrap), B5 (dispersion of the LPPLS critical time), B7 (Markov switching on the S&P 500), B9 (size under a volatility shift)

Proposed: you solve them

- Part A: A2 (Blanchard–Watson; model: A1), A4 (mildly explosive root; model: A3), A6 (filter and EM step; model: A5), A8 (hazard rate and damping; model: A7)
- Part B: B2 (SADF vs GSADF; model: B1), B4 (BET and BET-FI; model: B1, B3), B6 (testing the confidence indicator; model: B5, B3), B8 (bootstrap LR for regimes; model: B7), B10 (size under GARCH; model: B9)
- Part C: C1, is there an AI bubble?; C2, critique an AI answer on SADF and LPPLS

Setup: Python Environment and Data

-  **Open in Colab:** [chapter17_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`, `xlrd`
- Series: S&P Composite real price and dividend (monthly, 1871–2024, Robert J. Shiller); Nasdaq 100, S&P 500, BET, BET-FI; Bitcoin; NVIDIA, Cisco; closes up to 18 September 2026
- Weekly and monthly series: last close of the week or month; stocks adjusted for dividends and splits

Formulas Used Today

- Fundamental value (Gordon): $F = D_1/(r - g)$; rational bubble: $\mathbb{E}_t[B_{t+1}] = (1 + r)B_t$ Blanchard & Watson (1982)
- Right-tailed ADF: $\Delta y_t = a + b y_{t-1} + \varepsilon_t$; under a unit root $t_b \Rightarrow \frac{\frac{1}{2}(W(1)^2 - 1) - W(1) \int W}{\{\int W^2 - (\int W)^2\}^{1/2}}$; under $\rho_T = 1 + c/k_T$ it diverges Phillips & Magdalinos (2007)
- GSADF = $\sup_{r_2} \sup_{r_1} \text{ADF}_{r_1}^2$; BSADF $_{r_2} = \sup_{r_1} \text{ADF}_{r_1}^2$ Phillips, Shi & Yu (2015a)
- Markov switching: duration $1/(1 - p_{ii})$; stationary $\pi_2 = p_{12}/(p_{12} + p_{21})$; filter $P(s_t | \text{data}_t) \propto P(s_t | \text{data}_{t-1}) \varphi(r_t; \mu_{s_t}, \sigma_{s_t})$ Hamilton (1989)
- LPPLS: $\ln p = A + B(t_c - t)^m + (t_c - t)^m [C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))]$; $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2}$, $D = \frac{m|B|}{\omega|C|}$ Shu & Zhu (2020)

A1: Solving the Pricing Equation Forward [Solved]

Task

- Context: a share with price P_t and dividend D_t ; no-arbitrage $P_t = \mathbb{E}_t[P_{t+1} + D_{t+1}]/(1+r)$ with a constant required return $r > 0$; free disposal, $P_t \geq 0$ [Blanchard & Watson \(1982\)](#)
- (a) Substitute the equation into itself h times and write P_t as discounted expected dividends up to $t+h$ plus a remainder
- (b) Let $h \rightarrow \infty$: define F_t and B_t and prove that $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$
- (c) Prove that $B_t \geq 0$ and that $B_t = 0$ implies $B_{t+k} = 0$ for every $k \geq 1$ (Diba–Grossman) [Diba & Grossman \(1988\)](#)
- (d) Gordon case $D_1 = 2$, $g = 2\%$, $r = 6\%$, $P = 65$: compute F , B and the ratio $\mathbb{E}[B_h]/(F_h + \mathbb{E}[B_h])$ for $h = 5, 30, 100$ years
- Report: the derivation and the three ratios

Interpretation

- Why must a rational bubble exist from the first trade?
- What happens to the bubble share of the price in the long run, and why is this the testable implication?

A1: Solving the Pricing Equation Forward (Solution 1/2) [Solved]

Solution

- (a) Write $P_{t+1} = \mathbb{E}_{t+1}[P_{t+2} + D_{t+2}]/(1+r)$ and substitute; the law of iterated expectations gives $\mathbb{E}_t \mathbb{E}_{t+1} = \mathbb{E}_t$
- After h steps: $P_t = \sum_{i=1}^h \frac{\mathbb{E}_t D_{t+i}}{(1+r)^i} + \frac{\mathbb{E}_t P_{t+h}}{(1+r)^h}$
- (b) The sum converges to $F_t = \sum_{i \geq 1} \mathbb{E}_t D_{t+i}/(1+r)^i$; the remainder converges to $B_t = \lim_h \mathbb{E}_t P_{t+h}/(1+r)^h$, so $P_t = F_t + B_t$
- Shift one period and use iterated expectations:
 $\mathbb{E}_t B_{t+1} = \lim_h \mathbb{E}_t P_{t+1+h}/(1+r)^h = (1+r) \lim_h \mathbb{E}_t P_{t+h+1}/(1+r)^{h+1} = (1+r)B_t$
- Transversality, $\lim_h \mathbb{E}_t P_{t+h}/(1+r)^h = 0$, is exactly the condition $B_t = 0$

A1: Solving the Pricing Equation Forward (Solution 2/2) [Solved]

Solution

- (c) If $B_t < 0$, then $\mathbb{E}_t B_{t+h} = (1+r)^h B_t \rightarrow -\infty$ faster than $\mathbb{E}_t F_{t+h}$ can grow, so $\mathbb{E}_t P_{t+h} < 0$ for large h , which contradicts $P \geq 0$
- If $B_t = 0$: $\mathbb{E}_t B_{t+1} = 0$ and $B_{t+1} \geq 0$, so $B_{t+1} = 0$ almost surely; by induction $B_{t+k} = 0$ for all k
- (d) $F = D_1/(r-g) = 2/0.04 = 50.0$; $B = 65 - 50.0 = 15.0$, i.e. 23.1% of the price
- $F_h = 50.0 \cdot 1.02^h$, $\mathbb{E} B_h = 15.0 \cdot 1.06^h$: ratio 26.7% ($h = 5$), 48.8% ($h = 30$), 93.4% ($h = 100$)
- Interpretation: a bubble cannot be born later, and since $r > g$ it eventually dominates the price; relative to fundamentals the price path is explosive, which is what the right-tailed tests look for

A2: The Blanchard–Watson Bubble [Proposed]

Solution discussed in class

Task

- Context: $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π and $B_{t+1} = \varepsilon_{t+1}$ with probability $1 - \pi$, independent of the past; $\mathbb{E}\varepsilon = 0$; $B_0 = 15$, $r = 6\%$, $\pi = 0.9$ per year [Blanchard & Watson \(1982\)](#)
- (a) Derive $\mathbb{E}_t B_{t+1}$ and compare it with the martingale condition of A1
- (b) For $\varepsilon \equiv 0$, derive the conditional standard deviation $\text{sd}_t(B_{t+1})$ and evaluate it at $B_t = 15$
- (c) Give the distribution of the lifetime N , then $\mathbb{E}N$, $P(N > 5)$ and the growth rate while the bubble survives
- (d) In logs, $y = \ln B$, with a collapse sending y to a fixed level c : derive $\mathbb{E}[\Delta y_{t+1} \mid y_t]$ and its slope in y_t
- Model: A1

Interpretation

- Who pays for the crash risk, and how?
- What does a full-sample ADF regression on $\ln B$ see, and why is this the Evans pitfall?

A3: The Dickey–Fuller Limit with a Constant [Solved]

Task

- Context: $y_t = y_{t-1} + \varepsilon_t$, ε_t iid $(0, \sigma^2)$, $y_0 = 0$; regression $\Delta y_t = a + b y_{t-1} + e_t$ on $t = 1, \dots, T$; $t_b = \hat{b}/\text{SE}(\hat{b})$
- (a) Write $\hat{b}$ with the demeaned regressor $\tilde{y}_{t-1}$
- (b) Using the FCLT $T^{-1/2}y_{\lfloor Tr \rfloor} \Rightarrow \sigma W(r)$ and continuous mapping, find the limits of $T^{-2} \sum \tilde{y}_{t-1}^2$ and $T^{-1} \sum \tilde{y}_{t-1} \varepsilon_t$
- (c) Derive the limit of t_b ; (d) show that the limit numerator has mean $-1/2$
- (e) Simulate 20 000 random walks with $T = 1\,000$; report the 5%, 50%, 95% quantiles of t_b and the share of negative values

Interpretation

- Why is the 95% right-tail critical value negative instead of 1.645?
- What goes wrong if a bubble test uses 1.645?

A3: The Dickey–Fuller Limit with a Constant (Solution 1/2) [Solved]

Solution

- (a) OLS with a constant: $\hat{b} = \sum \tilde{y}_{t-1} \Delta y_t / \sum \tilde{y}_{t-1}^2 = \sum \tilde{y}_{t-1} \varepsilon_t / \sum \tilde{y}_{t-1}^2$ under H_0
- (b) $T^{-2} \sum \tilde{y}_{t-1}^2 \Rightarrow \sigma^2 \int_0^1 \tilde{W}^2$ with $\tilde{W} = W - \int_0^1 W$; $T^{-1} \sum y_{t-1} \varepsilon_t \Rightarrow \frac{\sigma^2}{2} (W(1)^2 - 1)$, and demeaning subtracts $\sigma^2 W(1) \int_0^1 W$
- (c) $SE(\hat{b}) = s / \sqrt{\sum \tilde{y}_{t-1}^2}$ with $s \rightarrow \sigma$, so σ cancels:
- $t_b \Rightarrow \frac{\frac{1}{2}(W(1)^2 - 1) - W(1) \int_0^1 W}{\{\int_0^1 W^2 - (\int_0^1 W)^2\}^{1/2}}$ (PSY, eq. 5 with $r_1 = 0$, $r_2 = 1$) Phillips, Shi & Yu (2015a)

A3: The Dickey–Fuller Limit with a Constant (Solution 2/2) [Solved]

Solution

- (d) $\mathbb{E}[W(1)^2] = 1$ and $\mathbb{E}[W(1) \int_0^1 W] = \int_0^1 \mathbb{E}[W(1)W(r)] dr = \int_0^1 r dr = \frac{1}{2}$, so the numerator has mean $\frac{1}{2}(1 - 1) - \frac{1}{2} = -\frac{1}{2}$ (simulated: -0.50)
- (e) 20 000 walks, $T = 1\,000$: quantiles 5%: -2.87 , 50%: -1.57 , 95%: -0.09 ; 96% of t_b are negative
- Interpretation: the estimated constant absorbs part of the random walk and pushes $\hat{b}$ down, so the distribution sits to the left; with 1.645 a right-tailed test would almost never reject: its size would be close to zero and its power very low

A4: A Mildly Explosive Root [Proposed]

Solution discussed in class

Task

- ▣ Context: $y_t = \rho_T y_{t-1} + \varepsilon_t$, $y_0 = 0$, ε_t iid $(0, \sigma^2)$, $\rho_T = 1 + c/k_T$ with $c > 0$, $k_T \rightarrow \infty$, $k_T = o(T)$ Phillips & Magdalinos (2007)
- ▣ (a) Write $y_t = \rho_T^t X_t$ with $X_t = \sum_{j \leq t} \rho_T^{-j} \varepsilon_j$ and find the limit of $\text{Var}(X_T)(\rho_T^2 - 1)$
- ▣ (b) Show that $\sum y_{t-1}^2 \approx X_T^2 \rho_T^{2T} / (\rho_T^2 - 1)$ and that $\frac{\rho_T^T}{\rho_T^2 - 1}(\hat{\rho} - \rho_T)$ is asymptotically a ratio of two independent Normal variables
- ▣ (c) Decompose the t -ratio of $b = \rho - 1$ and derive its divergence rate
- ▣ (d) Simulate 4 000 paths with $c = 1$, $k_T = T^{0.8}$, $T = 100, 400, 1\,600$ (regression with a constant); report ρ_T^T and the median t -ratio
- ▣ Model: A3

Interpretation

- ▣ Is the right-tailed test consistent against such bubbles? At what speed does its power grow?

A5: Markov Switching: Durations and One Filter Step [Solved]

Task

- Context: weekly returns in %, $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$; calm $\mu_1 = 0.3$, $\sigma_1 = 2$; turbulent $\mu_2 = -0.5$, $\sigma_2 = 5$; $p_{11} = 0.98$, $p_{22} = 0.90$ Hamilton (1989)
- (a) Compute the expected duration $1/(1 - p_{ii})$ of each regime
- (b) Compute the stationary probabilities $\pi_2 = p_{12}/(p_{12} + p_{21})$, $\pi_1 = 1 - \pi_2$
- (c) Compute the unconditional mean and standard deviation of r_t (law of total variance)
- (d) Last week $P(s = 2 \mid \text{data}) = 0.2$; this week $r_t = -6\%$: do one Hamilton filter step (prediction, densities, update)

Interpretation

- How much does one bad week change the regime probability, and why so much?

A5: Markov Switching: Durations and One Filter Step (Solution 1/2) [Solved]

Solution

- (a) Durations: $1/(1 - 0.98) = 50$ weeks (calm), $1/(1 - 0.90) = 10$ weeks (turbulent)
- (b) $\pi_2 = 0.02/(0.02 + 0.10) = 0.167$, $\pi_1 = 0.833$
- (c) $\mathbb{E}r = 0.833 \cdot 0.3 + 0.167 \cdot (-0.5) = 0.167$; $\text{Var}(r) = \sum_i \pi_i \sigma_i^2 + \sum_i \pi_i (\mu_i - \mathbb{E}r)^2 = 7.59$, s.d. 2.75

A5: Markov Switching: Durations and One Filter Step (Solution 2/2) [Solved]

Solution

- (d) Prediction: $P(s_t = 2 \mid \text{data}_{t-1}) = 0.2 \cdot 0.90 + 0.8 \cdot 0.02 = 0.196$
- Densities: $\varphi(-6; 0.3, 2^2) = 0.00140$, $\varphi(-6; -0.5, 5^2) = 0.04357$
- Update: $\frac{0.196 \cdot 0.04357}{0.196 \cdot 0.04357 + (1 - 0.196) \cdot 0.00140} = 0.884$
- Interpretation: -6% is more than three calm standard deviations below the calm mean, so its calm density is tiny; one such week almost settles the regime

A6: Another Filter Step and One EM Step [Proposed]

Solution discussed in class

Task

- ▣ Context: calm $\mu_1 = 0.4$, $\sigma_1 = 2$; turbulent $\mu_2 = -1$, $\sigma_2 = 6$; $p_{11} = 0.95$, $p_{22} = 0.80$; weekly returns in %
- ▣ (a) Compute durations, stationary probabilities, unconditional mean and standard deviation
- ▣ (b) Last week $P(\text{turbulent}) = 0.1$; this week $r_t = +1\%$: do one filter step
- ▣ (c) Compute the probability of the turbulent regime two weeks ahead, $(1 - p, p)P^2$
- ▣ (d) EM: given smoothed probabilities ξ_{tj} , maximise $\sum_t \sum_j \xi_{tj} \ln \varphi(r_t; \mu_j, \sigma_j^2)$ over μ_j and σ_j^2 [Hamilton \(1990\)](#)
- ▣ Model: A5

Interpretation

- ▣ Why does a small positive return point to the calm regime?
- ▣ Why is the M-step a weighted mean and variance?

A7: LPPLS: Slaving the Linear Parameters [Solved]

Task

- ▣ Context: $\ln p_t = A + Bf_t + C_1g_t + C_2h_t + u_t$ on a window $[t_1, t_2]$, with $f_t = (t_c - t)^m$, $g_t = f_t \cos(\omega \ln(t_c - t))$, $h_t = f_t \sin(\omega \ln(t_c - t))$ [Filimonov & Sornette \(2013\)](#)
- ▣ (a) For fixed (t_c, m, ω) , derive (A, B, C_1, C_2) in closed form
- ▣ (b) Write the profile objective that is left for the nonlinear search
- ▣ (c) Recover C and ϕ of $C \cos(\omega \ln(t_c - t) - \phi)$ from C_1, C_2
- ▣ (d) A fit on $[2025.0, 2026.0]$ gives $t_c = 2026.08$, $m = 0.5$, $\omega = 8$, $B = -1.2$, $C_1 = 0.06$, $C_2 = -0.02$: check every condition of Shu and Zhu (eq. 12) that the parameters alone allow [Shu & Zhu \(2020\)](#)

Interpretation

- ▣ Why does slaving make the calibration stable?
- ▣ Which conditions of the filter cannot be checked without the data?

A7: LPPLS: Slaving the Linear Parameters (Solution 1/2) [Solved]

Solution

- (a) With $X_t = (1, f_t, g_t, h_t)$ and $\beta = (A, B, C_1, C_2)^\top$: minimise $\sum_t (\ln p_t - X_t \beta)^2$
- Normal equations $(\sum_t X_t^\top X_t) \beta = \sum_t X_t^\top \ln p_t$: a 4×4 system whose entries are sums such as $N, \sum f_t, \sum f_t g_t, \sum h_t^2$;
 $\hat{\beta} = (X^\top X)^{-1} X^\top y$
- (b) Profile: $S(t_c, m, \omega) = y^\top y - y^\top X (X^\top X)^{-1} X^\top y$: only three nonlinear parameters, which removes most local minima of the seven-parameter search
- (c) $C \cos(x - \phi) = C \cos \phi \cos x + C \sin \phi \sin x$, so $C_1 = C \cos \phi$, $C_2 = C \sin \phi$: $C = \sqrt{C_1^2 + C_2^2} = 0.0632$,
 $\phi = \text{atan2}(C_2, C_1)$

A7: LPPLS: Slaving the Linear Parameters (Solution 2/2) [Solved]

Solution

- (d) $B = -1.2 < 0$; $m = 0.5 \in [0.01, 0.99]$; $\omega = 8 \in [2, 25]$: pass
- $t_c - t_2 = 0.08 \leq (t_2 - t_1)/5 = 0.2$, i.e. 29 days after the window: pass
- Oscillations (half-periods): $O = \frac{8}{\pi} \ln \frac{1.08}{0.08} = 6.63 \geq 2.5$: pass; damping $\frac{0.5 \cdot 1.2}{8 \cdot 0.0632} = 1.19 \geq 1$: pass
- The largest relative error, the Lomb test and the AR(1) test of the residuals need the price data: the fit is a candidate, not yet a qualified fit

A8: Why the Damping Condition? The Hazard Rate [Proposed]

Solution discussed in class

Task

- Context (JLS): before the crash, $\mathbb{E}_t[dp] = 0$ requires the drift $\mu(t) = \kappa h(t)$, with $h(t) \geq 0$ the crash hazard rate and $\kappa > 0$ the crash size; the log-price trend is $B\tau^m + C\tau^m \cos(\omega \ln \tau - \phi)$, $\tau = t_c - t$ [Johansen, Ledoit & Sornette \(2000\)](#)
- (a) Differentiate the trend with respect to t
- (b) Find the condition on (B, C, m, ω) for $h(t) \geq 0$ at every phase $\theta = \omega \ln \tau - \phi$
- (c) Show that for $\omega \gg m$ it becomes the damping condition $D = \frac{m|B|}{\omega|C|} \geq 1$ of Shu and Zhu (eq. 11) [Shu & Zhu \(2020\)](#)
- (d) Fit on [2025.0, 2026.0]: $t_c = 2026.45$, $m = 0.95$, $\omega = 3.5$, $B = -0.8$, $C_1 = 0.25$, $C_2 = 0$; check the exact condition, D , O and the t_c range
- Model: A7

Interpretation

- What would a negative hazard rate mean economically?
- Is the fit in (d) a bubble signature?

B1: GSADF on the S&P 500 Price-Dividend Ratio [Solved]

Task

- Data: S&P Composite real price and real dividend, monthly 1871–2024 (Robert J. Shiller); $y_t = \ln(P_t/D_t)$, $T = 1865$
- (a) Compute GSADF with a constant, no lags and $r_0 = 0.01 + 1.8/\sqrt{T}$; obtain the 95% critical value from 1000 random walks $y_t = T^{-1} + y_{t-1} + \varepsilon_t$ Phillips, Shi & Yu (2015a)
- (b) Repeat (a) with $r_0 = 0.03$ and $r_0 = 0.08$
- (c) For each r_0 , date-stamp with BSADF against its pointwise 95% critical values, minimum duration $\lceil \ln T \rceil$ months; mark each episode as a rise or a fall of the ratio
- Report: a table r_0 , w_0 , GSADF, critical value, episodes

Interpretation

- Which episodes survive every window choice?
- Why can a very short minimum window produce spurious episodes?
- Is every explosive episode of the ratio a bubble?

B1: GSADF on the S&P 500 Price-Dividend Ratio (Solution 1/2) [Solved]

r_0	w_0	GSADF	cv 95%	Episodes (BSADF)
0.030	55	3.93	2.55	Jan 1879–Apr 1880; Aug 1917–May 1918 (falling); Nov 1928–Sep 1929; Dec 1954–Jan 1956; May 1997–Aug 2001; Sep 2008–Apr 2009 (falling)
0.052	96	3.41	2.39	Aug 1917–Jun 1918 (falling); Nov 1928–Sep 1929; May 1997–Aug 2001
0.080	149	1.77	2.24	May 1997–Aug 2001

- Steps: log ratio; for every end date, ADF on all windows of at least w_0 months; the supremum over windows gives BSADF, and over end dates GSADF; the same on the 1 000 simulated walks gives the critical values

B1: GSADF on the S&P 500 Price-Dividend Ratio (Solution 2/2) [Solved]

Solution

- The dot-com episode (1997–2001) is date-stamped with every window; with $r_0 = 0.08$ GSADF is below its critical value, yet BSADF still crosses its pointwise critical values: the global test is stricter than the date-stamping
- The shortest window adds 1878–1880, 1928–1929, 1955–1956 and 2008–2009: fewer observations per regression, noisier ADF statistics, more short crossings
- 1917–1918 is a fall of the ratio (real prices dropped faster than real dividends): explosive, but not a bubble
- Interpretation: only episodes that survive every r_0 and point upwards support the bubble reading

B2: SADF vs GSADF [Proposed]

Solution discussed in class

Task

- Data: monthly log closes of the Nasdaq 100 and of the S&P 500, 1990–2026 (last close of each month)
- (a) Compute SADF and GSADF with $r_0 = 0.01 + 1.8/\sqrt{T}$ and their 95% critical values from 2 000 simulated random walks
- (b) Date-stamp with the recursive ADF of PWY (window start fixed at the first month) and with BSADF of PSY, minimum duration $\lceil \ln T \rceil$ months [Phillips, Wu & Yu \(2011\)](#)
- (c) List the episodes of both methods with their direction (rise or fall)
- Model: B1

Interpretation

- Which episodes does only one of the two methods find, and why?
- Is any episode an accelerating fall rather than a rise?

B3: Bitcoin: Simulated vs Bootstrap Critical Values [Solved]

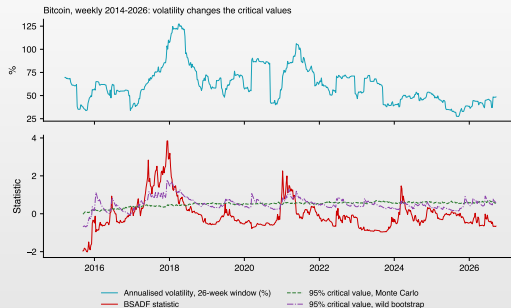
Task

- Data: weekly log closes of Bitcoin, September 2014–September 2026 ($T = 626$ weekly changes)
- (a) Compute GSADF ($r_0 = 0.01 + 1.8/\sqrt{T}$, constant, no lags) with Monte Carlo critical values from 2 000 Gaussian random walks
- (b) Wild bootstrap: multiply the demeaned weekly changes by independent random signs $w_t = \pm 1$, rebuild the path, recompute BSADF and GSADF; 1 000 replications; take the 95% quantiles [Harvey et al. \(2016\)](#)
- (c) Date-stamp with both critical-value sequences, minimum duration $\lceil \ln T \rceil$ weeks; report GSADF, both critical values, the episodes and a chart of BSADF with both sequences

Interpretation

- When are the bootstrap critical values higher, and why?
- Which episode is robust to the choice of critical values?

B3: Bitcoin: Simulated vs Bootstrap Critical Values (Solution 1/2) [Solved]



B3: Bitcoin: Simulated vs Bootstrap Critical Values (Solution 2/2) [Solved]

Solution

- (a)–(b) GSADF = 3.85; 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82; $3.85 > 2.82$: reject the random walk at 5% with either
- (c) Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- The bootstrap critical values rise in and after the volatile periods (maximum 1.80 vs 0.71 for Monte Carlo), because the resampled changes keep the large shocks where they occurred
- Interpretation: the 2017 episode survives both, shorter with the bootstrap; the 2021 episode shrinks: its explosiveness is partly a volatility effect

B4: The Bucharest Stock Exchange [Proposed]

Solution discussed in class

Task

- Data: official BET index, monthly log closes 1997–2026; BET-FI (financial investment companies), weekly log closes 2012–2026
- (a) For each index, compute GSADF with Monte Carlo (2000 paths) and wild bootstrap (1000 replications) 95% critical values
- (b) Date-stamp with both critical-value sequences (minimum duration $\lceil \ln T \rceil$) and classify each episode as a rise or a fall
- (c) Report a table (T , GSADF, both critical values) and the list of episodes
- Model: B1, B3

Interpretation

- Was the 2003–2007 boom explosive in real time? When did the signal start? [Pele, Mazurencu & Nijkamp \(2013\)](#)
- Is the current rise of both indices explosive, and does that make it a bubble?

B5: How Much Does the Critical Time Depend on the Window? [Solved]

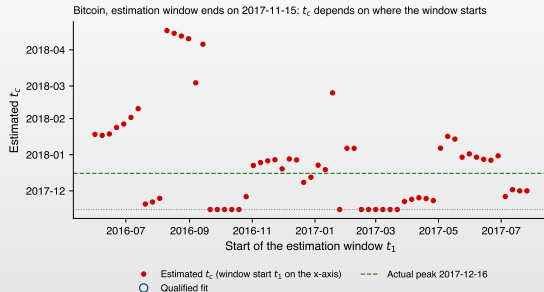
Task

- Data: Bitcoin daily log closes; all windows end on $t_2 = 15$ November 2017 and start every 7 days between 1 June 2016 and 1 August 2017 (61 windows)
- (a) Fit LPPLS on each window: (A, B, C_1, C_2) by OLS, (t_c, m, ω) by grid and Nelder–Mead within the search space of Shu and Zhu (eq. 11) [Filimonov & Sornette \(2013\)](#), [Shu & Zhu \(2020\)](#)
- (b) Report the minimum, median and maximum t_c , the interquartile range of t_c minus the actual peak, the share of qualified fits (eq. 12), the share at the edge of the search space and the share within 30 days of the peak

Interpretation

- Would you trust a single t_c reported without its window?
- What does a fit at the edge of the search space tell you?

B5: How Much Does the Critical Time Depend on the Window? (Solution 1/2) [Solved]



B5: How Much Does the Critical Time Depend on the Window? (Solution 2/2) [Solved]

Solution

- t_c ranges from 15 Nov 2017 to 17 Apr 2018, median 23 Dec 2017; actual peak 16 Dec 2017; interquartile range of the error 53 days, median error 7 days
- Qualified fits: 0%; fits at the edge of the ranges of m or ω : 43%; within 30 days of the peak: 56%
- A fit at the edge of the search space is not an interior optimum: the data do not pin the parameter down
- Interpretation: the same end date gives critical times months apart, depending only on the start of the window; a single t_c is not a forecast

B6: Does the Confidence Indicator Beat Chance? [Proposed]

Solution discussed in class

Task

- Data: Nasdaq 100 daily closes; LPPLS confidence indicator from 1997 on the grid of Shu and Zhu: 141 windows of 750 to 50 trading days, t_2 every 5 trading days [Shu & Zhu \(2020\)](#)
- (a) Find when the indicator was first positive and its share of positive dates between January 1999 and the March 2000 peak
- (b) Estimate $\Delta = P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) - P(\cdot \mid \text{no signal})$; build a 95% CI by a circular block bootstrap (blocks of 90 days, 9999 replications) and test $b = 0$ in a linear probability regression with Newey–West standard errors
- (c) Compare the block-bootstrap SE with the independent-data SE; compute the design effect
- (d) Power: how many independent observations detect +10 points (5% two-sided, power 80%), and how many dates is that here?
- Model: B5 (LPPLS), B3 (bootstrap)

Interpretation

- Would the indicator have helped an investor in 2000?
- Is the absence of evidence here evidence of absence?

B7: Markov Switching on the S&P 500 [Solved]

Task

- Data: weekly log returns of the S&P 500 in %, 1990–2026 (1 915 weeks)
- (a) Estimate a two-regime model $r_t = \mu_{s_t} + \sigma_{s_t} \varepsilon_t$ by maximum likelihood (Hamilton filter, 20 random starts); report estimates, standard errors and expected durations [Hamilton \(1989\)](#)
- (b) Find the first week of 2000, of autumn 2008 and of 2020 with a filtered turbulent probability above 0.5
- (c) Compute the share of weeks in which the filtered and the smoothed regime (above 0.5) agree
- (d) Test $\mu_{\text{turb}} = 0$ with a t -test at 5%, and compute $\text{LR} = 2(\hat{\ell}_2 - \hat{\ell}_1)$ against one Normal regime

Interpretation

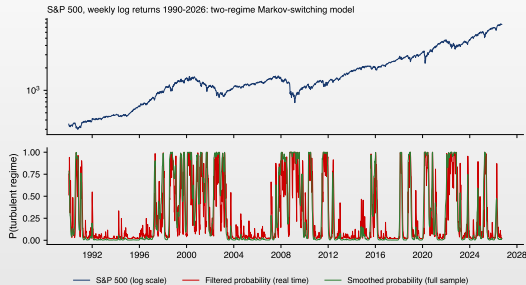
- Is the turbulent mean significantly negative?
- Why can LR not be compared with a χ^2 critical value?

B7: Markov Switching on the S&P 500 (Solution 1/2) [Solved]

Regime	$\hat{\mu}$	SE	$\hat{\sigma}$	$\hat{p}_{ii}$	Duration
Calm	0.32	0.04	1.49	0.974	38.1
Turbulent	-0.27	0.17	3.67	0.931	14.4

- (b) First week above 0.5: 28 Jan 2000 (2000), 3 Oct 2008 (autumn 2008), 28 Feb 2020 (2020)
- (c) Filtered and smoothed regimes agree in 90% of weeks

B7: Markov Switching on the S&P 500 (Solution 2/2) [Solved]



- (d) $t = -0.27/0.17$: $|t| < 1.96$, do not reject $\mu_{\text{turb}} = 0$ at 5%; $LR = 2(-4070.0 - (-4326.0)) = 512.1$, but the second regime is not identified under one regime, so χ^2 does not apply (B8)

B8: Bitcoin: Two or Three Regimes? [Proposed]

Solution discussed in class

Task

- Data: weekly log returns of Bitcoin in %, 2014–2026 (626 weeks)
- (a) Estimate two and three regimes with switching mean and variance (30 random starts); report log-likelihood, AIC, BIC and each regime's mean, s.d. and duration
- (b) Test two vs three regimes with $LR = 2(\hat{\ell}_3 - \hat{\ell}_2)$: simulate 199 samples of 626 weeks from the fitted two-regime model, refit both models on each, and compute the bootstrap p -value $(1 + \#\{LR^* \geq LR\})/(B + 1)$ [Hansen \(1992\)](#)
- (c) Compare the bootstrap 95% value with the χ^2_6 value
- Model: B7

Interpretation

- Which regime do the 2017 and 2021 booms belong to?
- Why is χ^2 the wrong reference distribution?

B9: Size of GSADF Under a Volatility Shift [Solved]

Task

- Data: simulated. 1 000 random walks $y_t = T^{-1} + y_{t-1} + \sigma_t z_t$, $z_t \sim N(0, 1)$, $T = 400$, $w_0 = 40$; $\sigma_t = 1$ up to $T/2$, then $\sigma_t = 3$: no bubble [Harvey et al. \(2016\)](#)
- (a) Compute the rejection rate of GSADF at 5% with the Monte Carlo critical value for constant variance (2.19)
- (b) Repeat with a wild bootstrap critical value computed for each path (199 replications, as in B3)
- (c) Compute the share of paths with at least one false date-stamped episode (BSADF above its pointwise 95% value for $\lceil \ln T \rceil$ periods)
- (d) Do the same for constant volatility as a control; report a table

Interpretation

- Which critical value would you trust for Bitcoin, and why?
- Does the wild bootstrap solve the multiplicity of date-stamping?

B9: Size of GSADF Under a Volatility Shift (Solution 1/2) [Solved]

	GSADF, MC cv	GSADF, wild cv	false episode, MC	false episode, wild
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%

- Steps: simulate the path; compute GSADF and BSADF; compare with the fixed Monte Carlo values and with the path's own wild bootstrap values; average the rejections over the 1 000 paths
- Monte Carlo error of a rejection rate near 5%: ± 1.4 points (95%)

B9: Size of GSADF Under a Volatility Shift (Solution 2/2) [Solved]

Solution

- (a) The late volatility rise looks like acceleration: 32.0% false rejections with the constant-variance critical value, against 4.1% in the control
- (b) The wild bootstrap keeps the variance path of each sample and restores 4.1%
- (c) False episodes occur in about half of the paths or more with either critical value: the bootstrap fixes the size of GSADF, not the multiplicity of date-stamping (Lecture 17: family-wise critical values)
- Interpretation: for Bitcoin, whose volatility changes strongly, use the wild bootstrap for the global test and family-wise critical values for the dates

B10: Size Under GARCH Volatility [Proposed]

Solution discussed in class

Task

- Data: simulated, as in B9, with two new null models: GARCH(1,1) shocks $\sigma_t^2 = 0.02 + 0.10 e_{t-1}^2 + 0.88 \sigma_{t-1}^2$ (unconditional variance 1), and a volatility **fall** from 3 to 1 at $T/2$
- (a) Compute the size of GSADF at 5% with Monte Carlo and with wild bootstrap critical values (1 000 paths, 199 replications each)
- (b) Compute the share of paths with a false date-stamped episode; report a table as in B9
- Model: B9

Interpretation

- Why does a fall in volatility not create false bubbles, while a rise does?
- Is the wild bootstrap exact under GARCH? What does it keep, and what not?

C1: Is There an AI Bubble?

- Compare NVIDIA and the Nasdaq 100 today with Cisco and the Nasdaq 100 in the dot-com episode
- Explosiveness: GSADF/BSADF on monthly and weekly data, Monte Carlo and wild bootstrap critical values
 - ▶ When did each signal start; how far had prices risen by then; is the current episode still open?
- Run-ups à la Greenwood, Shleifer and You: two-year return, raw and net of the Nasdaq 100 [Greenwood, Shleifer & You \(2019\)](#)
- LPPLS: is there a robust super-exponential signature today (confidence indicator)?
- Fundamentals: which data on earnings or sales would you add to separate a bubble from explosive fundamentals [Pástor & Veronesi \(2009\)](#)?
- Required: test explosiveness of the log price-to-sales or price-to-earnings ratio, not only of prices, and date-stamp with family-wise critical values [Phillips & Shi \(2020\)](#)
- Deliverable: a two-page note with charts, and a Quantlet folder (code, Metainfo, charts)

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **Student prompt:** “y is the monthly log price-dividend ratio of the S&P 500. Write Python code for SADF with the PSY minimum window, decide at 5% whether there is a bubble, date-stamp it and tell me when it will burst.”

- **Answer of the AI assistant**

```
import numpy as np
from statsmodels.tsa.stattools import adfuller
T = len(y)
w0 = int(np.floor((0.01 + 1.8 / np.sqrt(T)) * T))
adf = lambda x: adfuller(x, maxlag=0, regression='c', autolag=None)[0]
sadf = max(adf(y[:t]) for t in range(w0, T + 1))
cv = adfuller(y, maxlag=0, regression='c', autolag=None)[4]['5%']
print('bubble' if sadf > cv else 'no bubble')
flag = [adf(y[t - w0:t + w0]) > cv for t in range(w0, T - w0)]
```

- “An LPPLS fit on the last 36 months gives $\hat{t}_c = \text{March 2027}$: this is when the bubble bursts.”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

- The answer on the previous slide contains **three** errors typical for bubble tests; the rest is correct
 - ▶ SADF: supremum of ADF statistics over forward-expanding windows (Lecture 17)
 - ▶ PSY: Phillips, Shi and Yu, the authors of the GSADF test [Phillips, Shi & Yu \(2015a\)](#)
- Your tasks
 - ▶ Find each error: which line, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a test on simulated data, a truncated sample, a second window
 - ▶ Give the corrected code or statement
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls

- ▣ Using the left-tailed Dickey–Fuller table for a right-tailed test
- ▣ Calling an explosive episode a bubble without checking its direction (up or down)
- ▣ Reading date-stamps from smoothed or full-sample quantities as if they were available in real time
- ▣ Reporting one LPPLS critical time without the window, the filter and its dispersion
- ▣ Ignoring volatility shifts in crypto data: use the wild bootstrap
- ▣ Using weekly or monthly data for short squeezes that last days

Take-Home Assignment (Team Project)

- Choose three assets not covered in class (a BVB stock, a crypto-asset, a sector or country index)
- Run GSADF with Monte Carlo and wild bootstrap critical values at two frequencies; date-stamp and classify the episodes
- Fit LPPLS on the largest episode; report the dispersion of t_c across windows and the confidence indicator
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- If you use an AI assistant, log every error you catch in [AI_ERRORS.md](#) (as in C2)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Cryptocurrency: Speculative Asset and Medium of Exchange
- ▣  Statistics of Financial Markets

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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